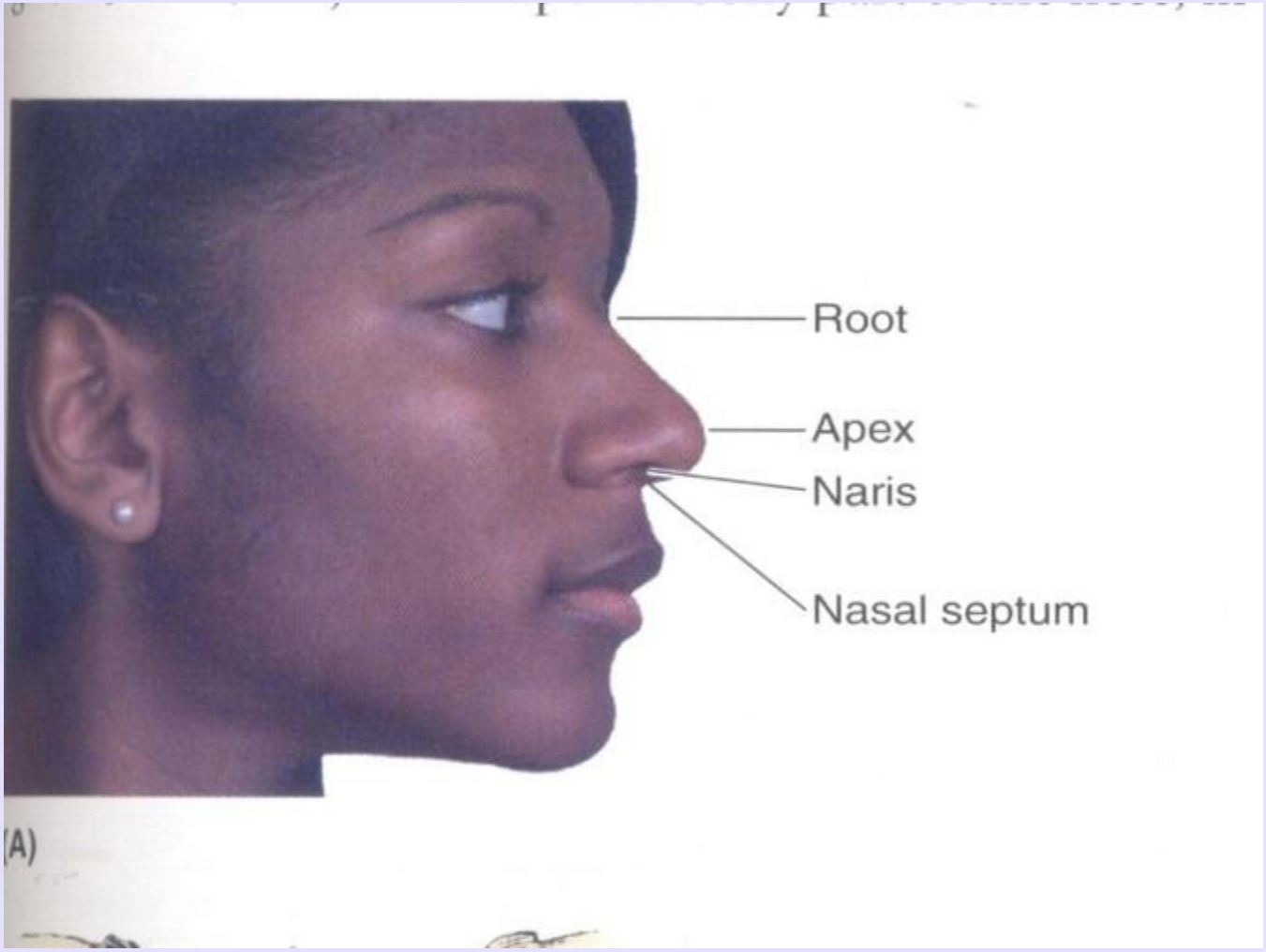
EXTERNAL NOSE & BONY BOUNDARIES OF NASAL CAVITY.

Learning Objectives.

- At the end of lecture, students should be able to:
- Know the anatomy of external nose.
- Define the Boundaries of nasal cavity.
- Know Blood vessels of nose.
- Know Nerve supply of nose.

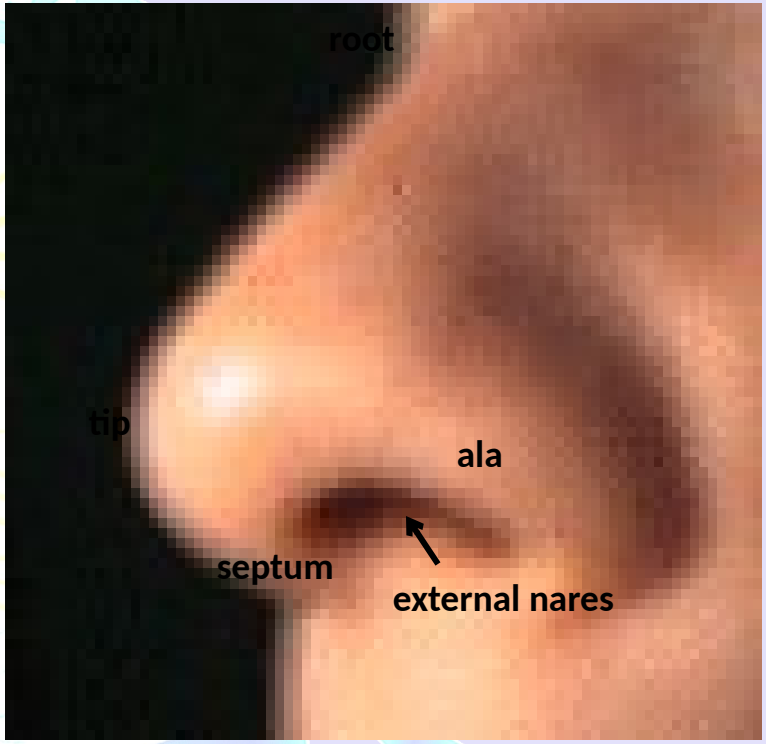
- **Nose:**
  - Organ of respiratory passage.
  - Organ of smell
  - Receptors for smell lies in upper 1/3rd of nose covered by olfactory mucosa and or factory cells.
  - Rest of nasal cavity lined by respiratory mucosa.

- **Functions :**
- serous glands to make the air moist
- mucous glands to trap the dust and foreign particles.


Root
Apex
Naris
Nasal septum
(A)

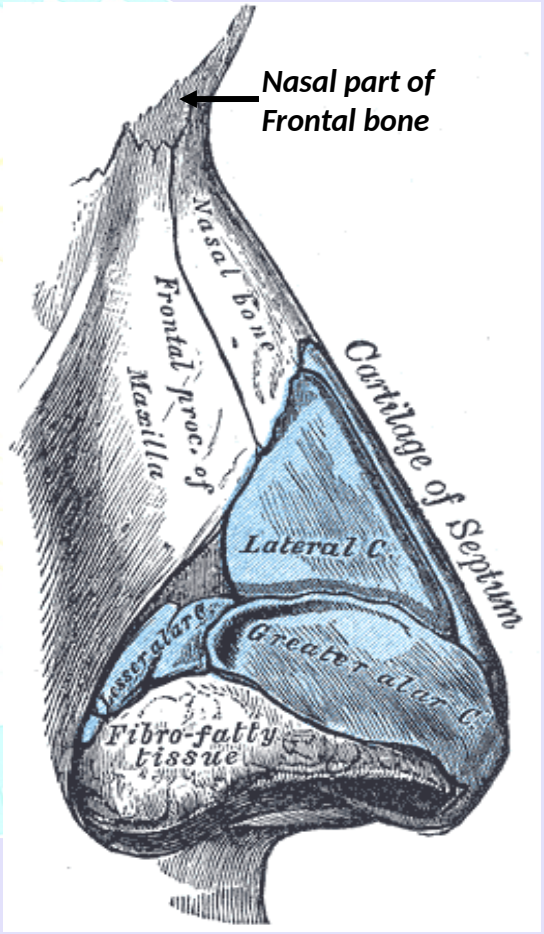
Nose

- Only externally visible part of the respiratory system
- Has a free tip and is attached to the forehead by the root or the bridge
- Has two openings, the anterior (external) nares or nostrils, which lead to the nasal cavity
- Each nostril is bounded laterally by the ala and medially by the nasal septum



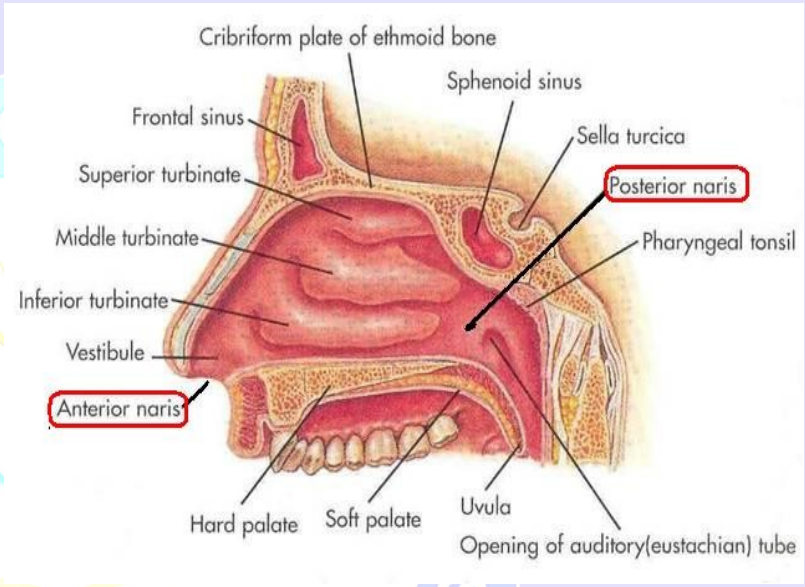
Nose: Structure

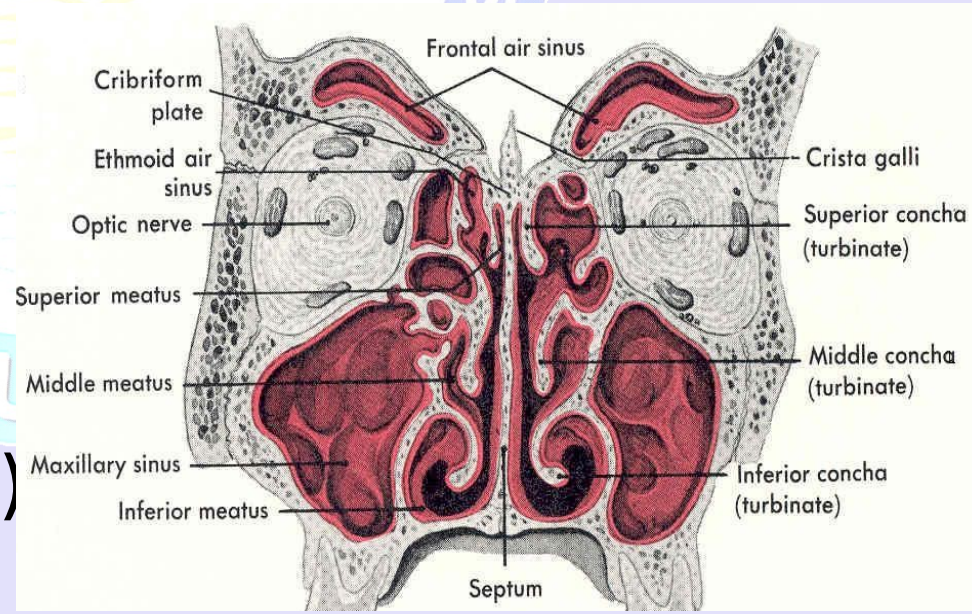
- Nose consists of bony & cartilaginous framework
- Formed above by the:
  - Nasal bones
  - Frontal processes of maxillae
  - Nasal part of frontal bone
- Formed below by plates of hyaline cartilage, which include upper & lower nasal cartilages and the septal cartilage



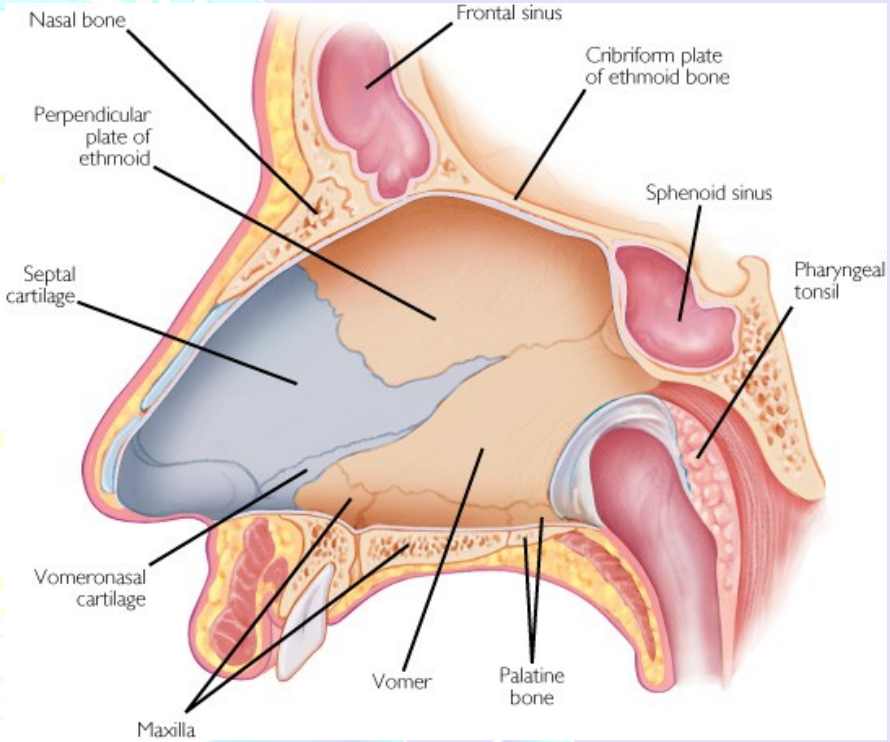
Nasal Cavity

- Extends from the external (anterior) nares to the posterior nares (choanae)
- Divided into right & left halves by the nasal septum
- Each half has a:
  - Floor
  - Roof
  - Lateral wall
  - Medial wall (septum)



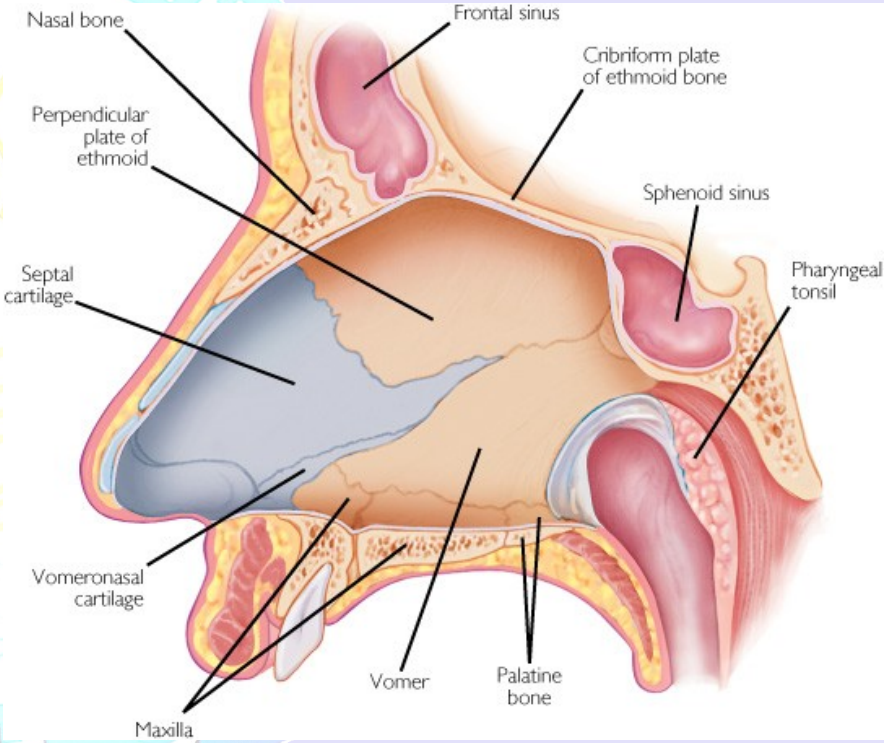


- Roof
- Is narrow & formed (from behind forward) by the:
  - Body of sphenoid
  - Cribriform plate of ethmoid bone
  - Frontal bone
  - Nasal bone & cartilage



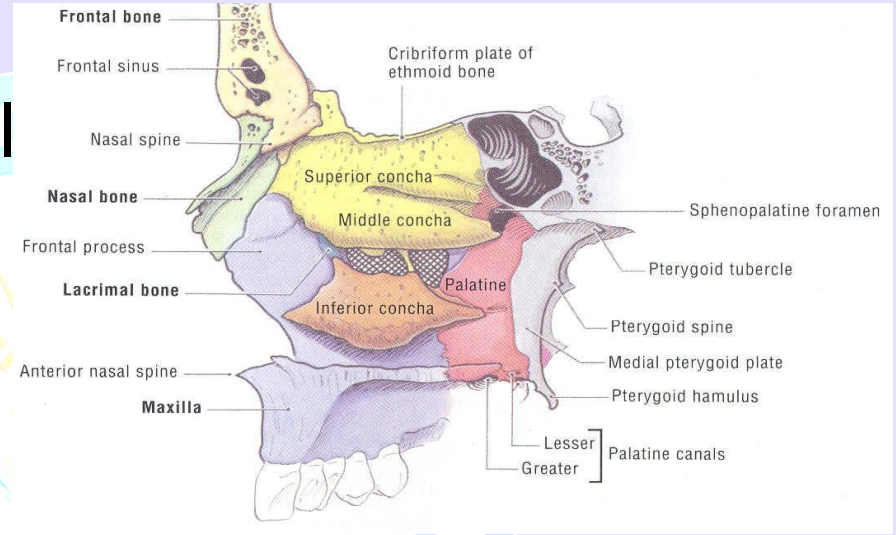
- Floor
  - Separates it from the oral cavity
  - Formed by the hard (bony) palate

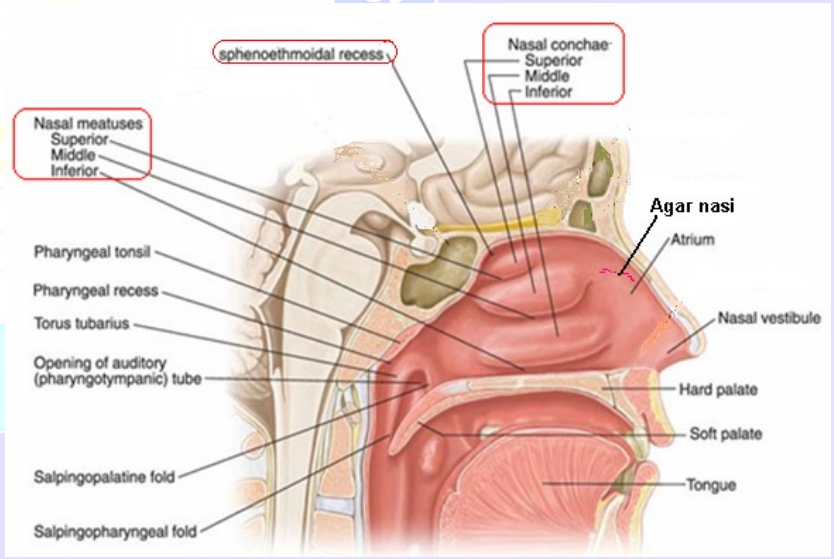
- Medial Wall (Nasal Septum)
- Osteocartilaginous partition, only rarely lying in the midline
- Covered by the mucoperiosteum
- Formed:
  - Superiorly by the vertical (perpendicular) plate of ethmoid bone
  - Posteriorly by the vomer bone
  - Anteriorly by the septal cartilage

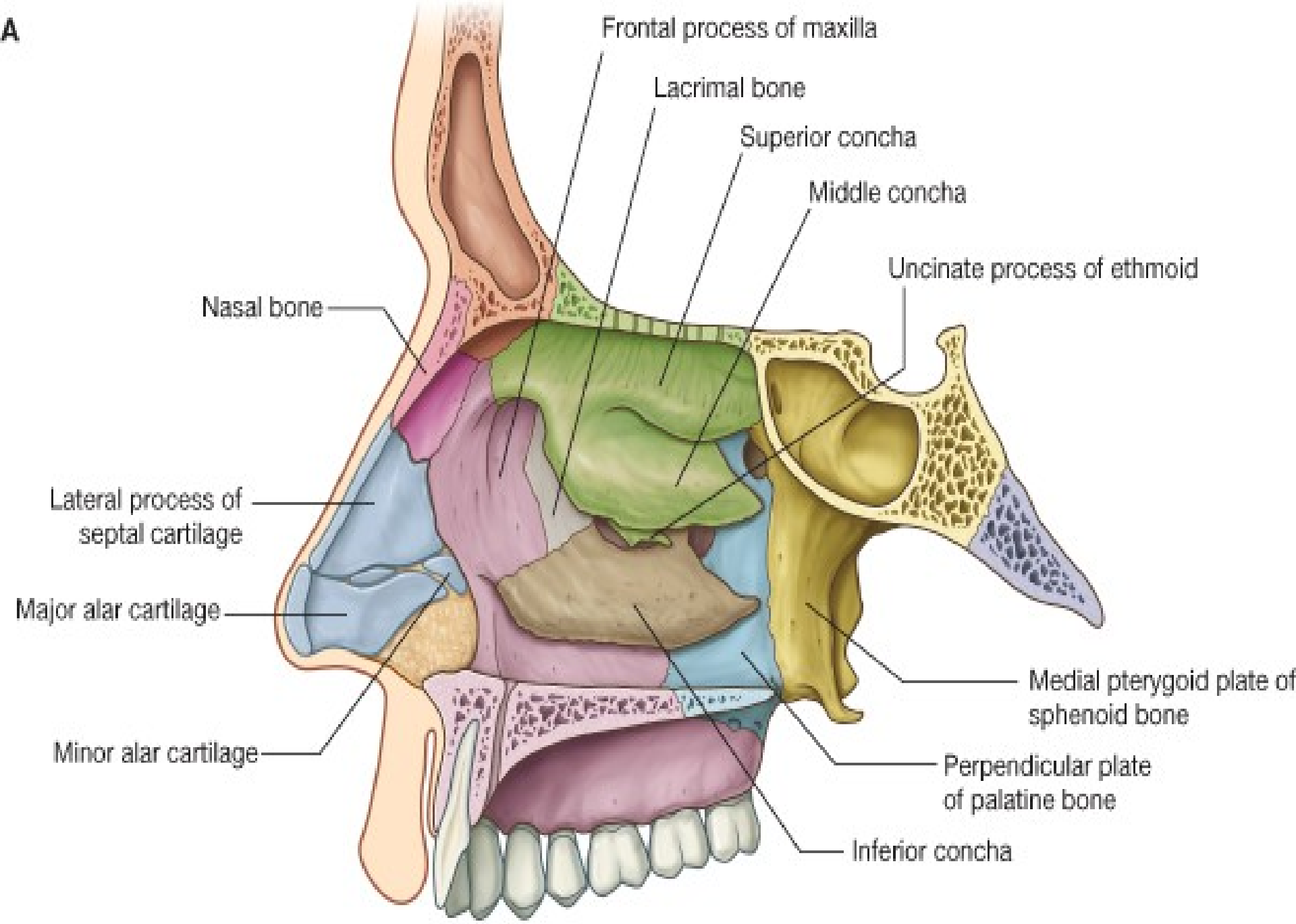


- Lateral Wall

- Shows three horizontal bony projections, covered by mucous membrane, the superior, middle & inferior conchae (turbinates)

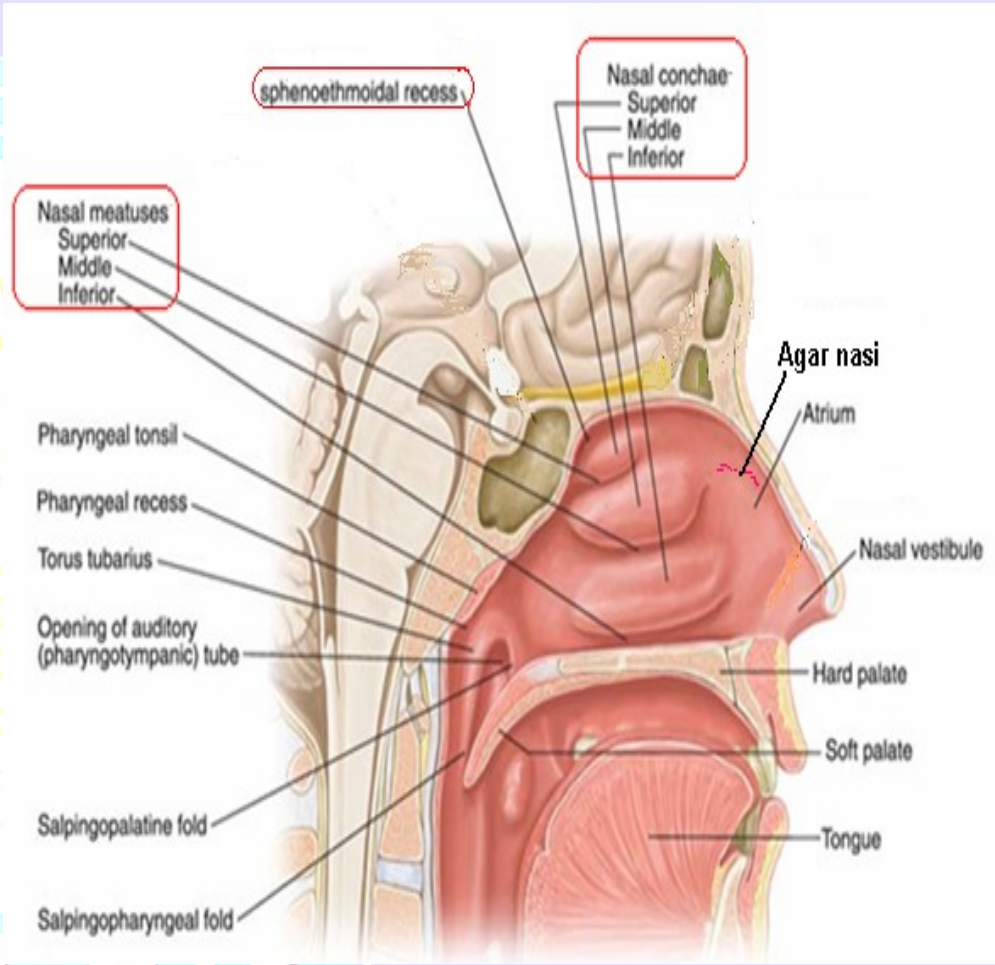




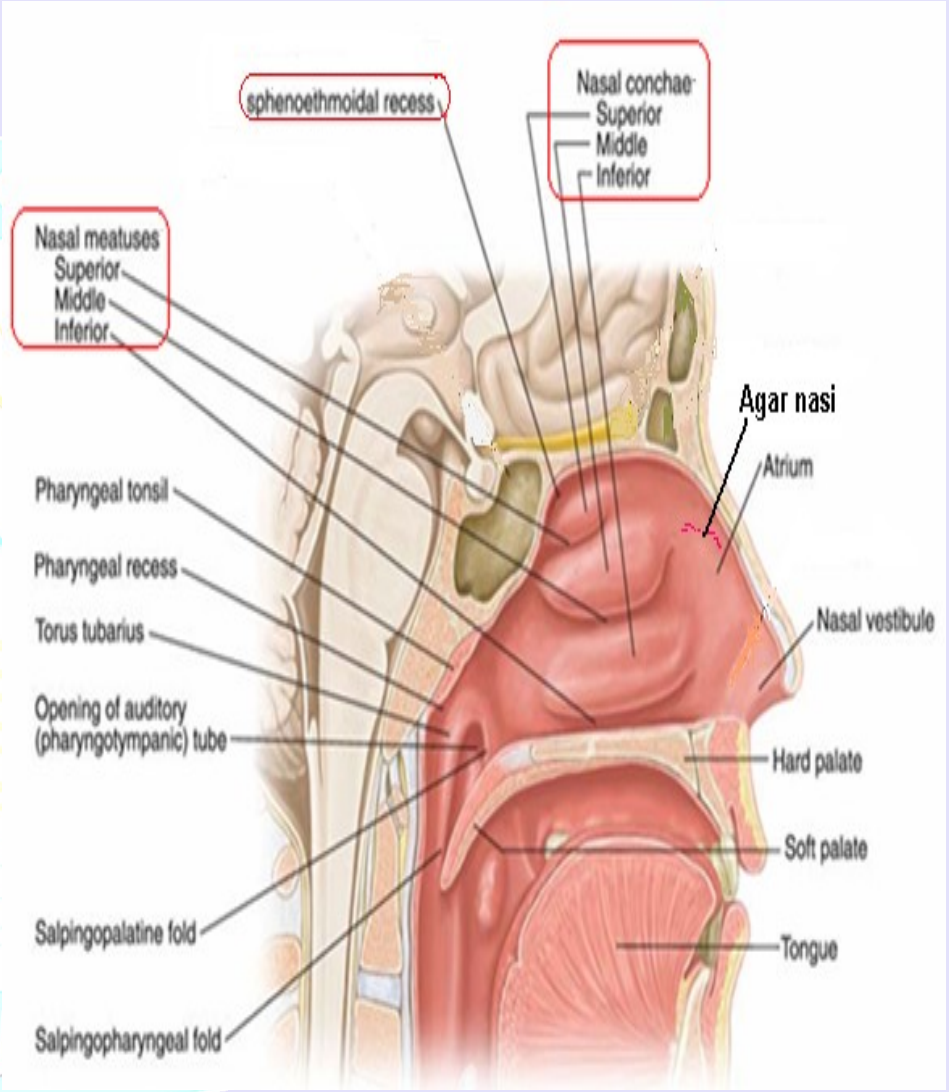

A
Frontal process of maxilla
Lacrimal bone
Superior concha
Middle concha
Uncinate process of ethmoid
Nasal bone
Lateral process of septal cartilage
Major alar cartilage
Minor alar cartilage
Medial pterygoid plate of sphenoid bone
Perpendicular plate of palatine bone
Inferior concha

- The superior and middle conchae are parts of the ethmoid bone, whereas the inferior concha is a separate bone.

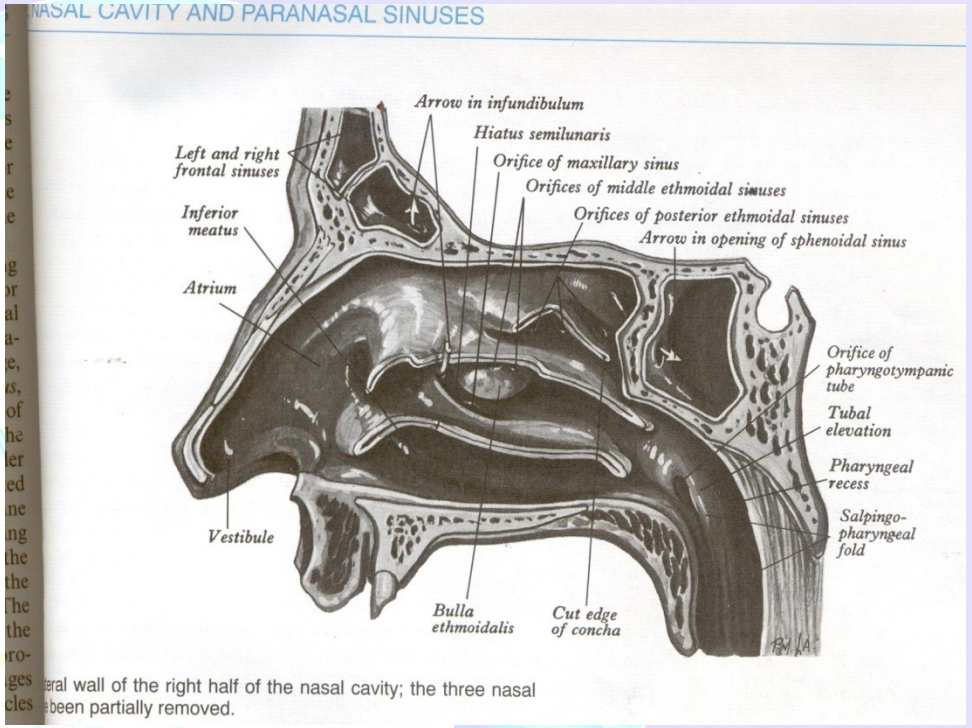
- The cavity below each concha is called a meatus and are named as superior, middle & inferior corresponding to the conchae



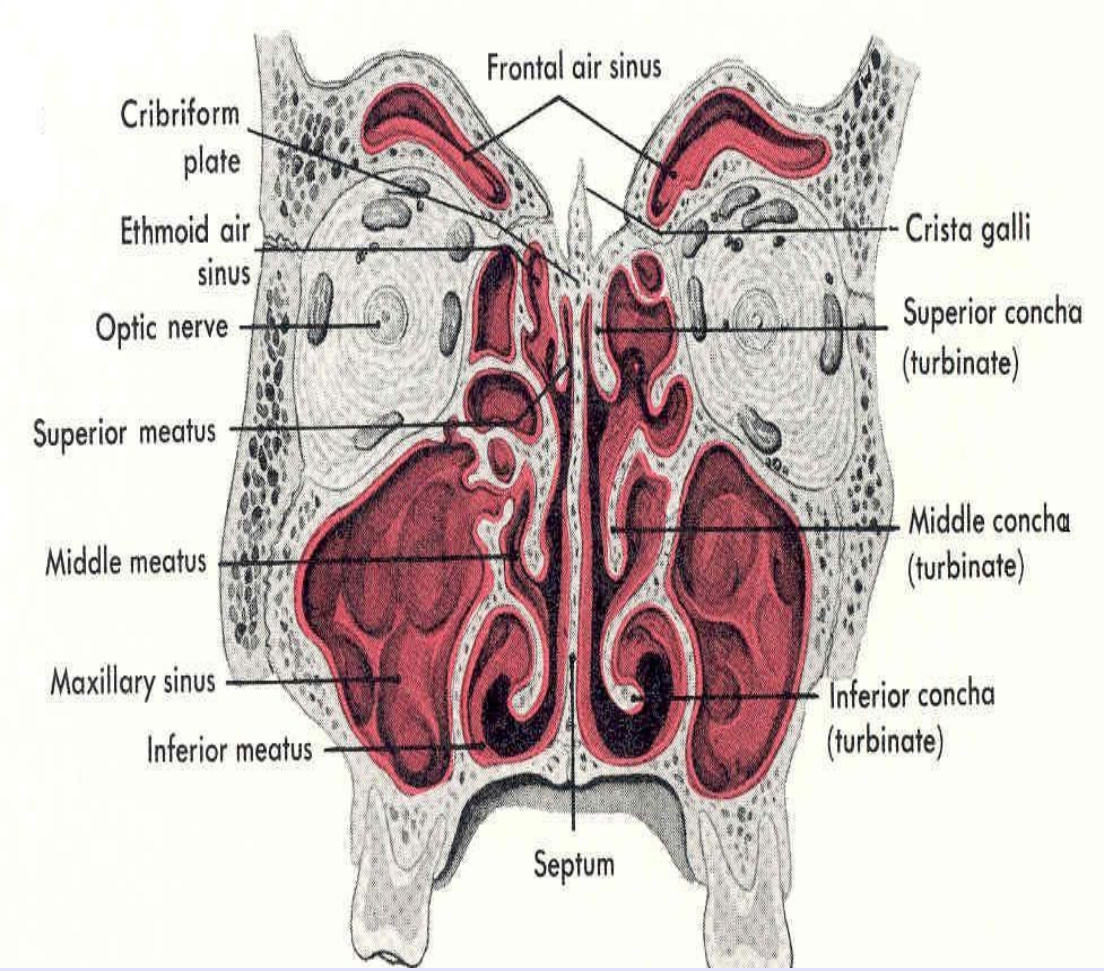
- The small space above the superior concha is called the sphenoethmoidal recess
- The middle meatus is continuous in front with a depression called the atrium

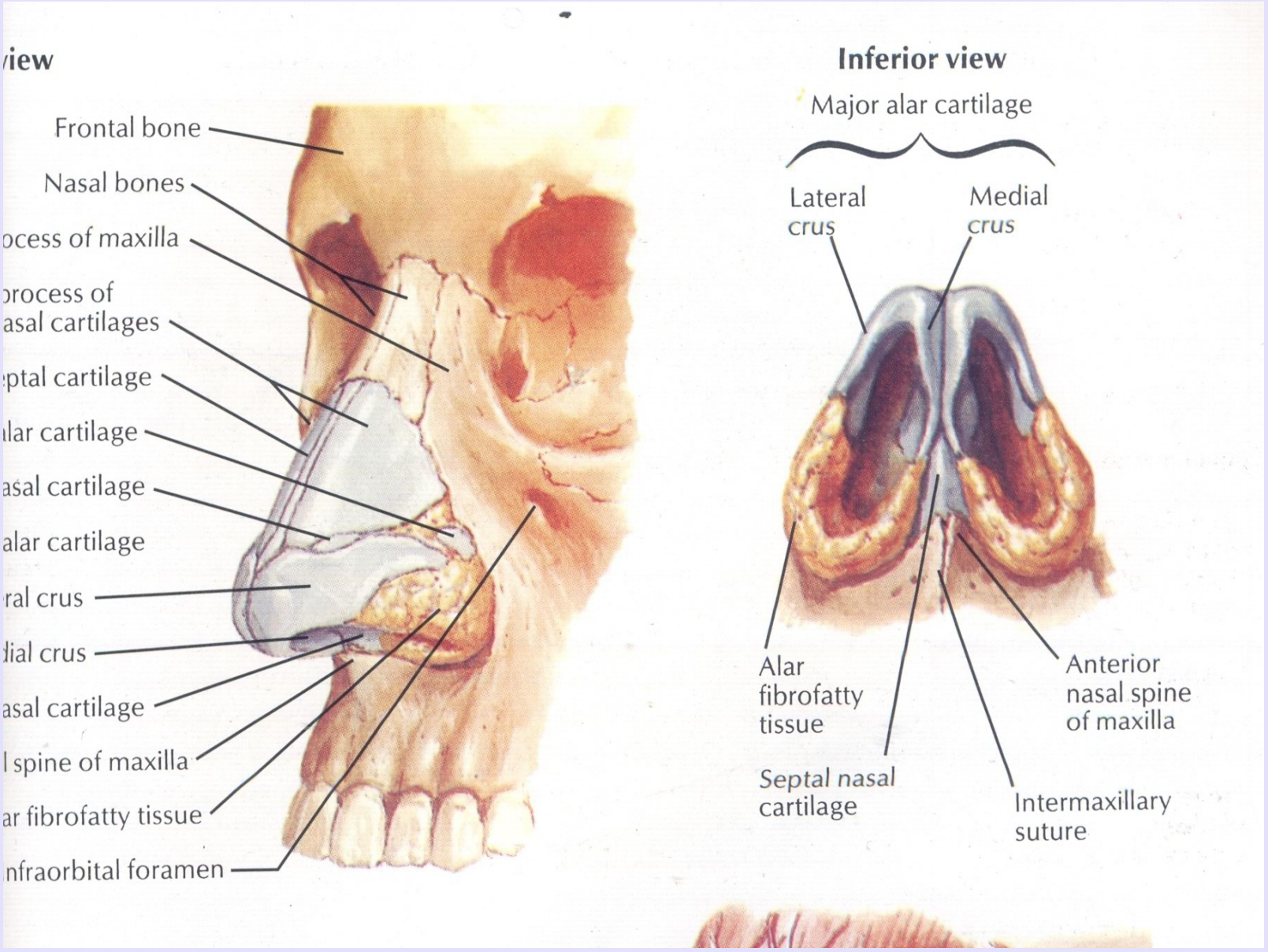


- Atrium is limited bove by a ridge called agar nasi
- Below and in front of atrium, and just within the nostril lies the vestibule



- The conchae increase the surface area of the nasal cavity
- The recess & meati receive the openings of the:
  - Paranasal sinuses
  - Nasolacrimal duct

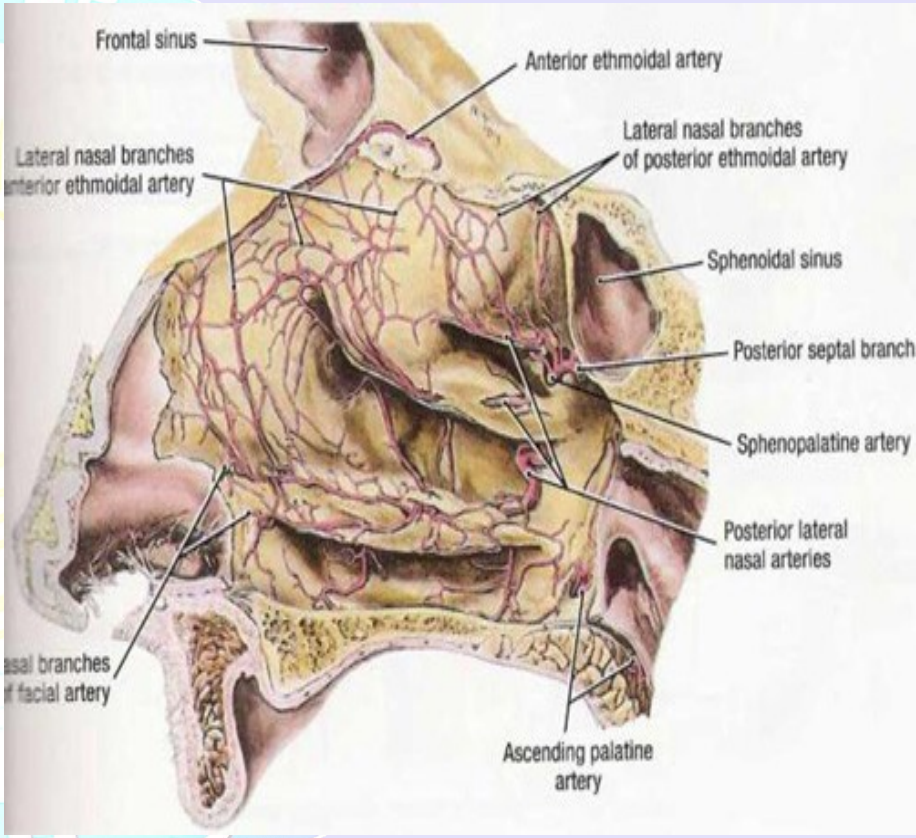



view
Frontal bone
Nasal bones
ocess of maxilla
process of
asal cartilages
ptal cartilage
lar cartilage
asal cartilage
alar cartilage
ral crus
dial crus
asal cartilage
l spine of maxilla
ar fibrofatty tissue
nfraorbital foramen
Inferior view
Major alar cartilage
Lateral
crus
Medial
crus
Alar
fibrofatty
tissue
Septal nasal
cartilage
Anterior
nasal spine
of maxilla
Intermaxillary
suture

Arterial supply of nose.

- The arterial supply of medial and lateral walls of nasal cavity is from five sources:

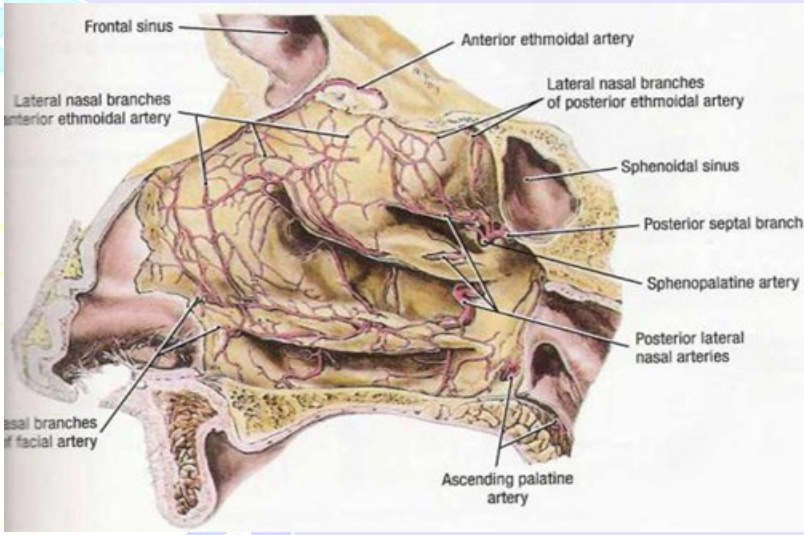
1. Anterior ethmoidal artery (from ophthalmic artery).
2. Posterior ethmoidal artery (from ophthalmic artery).
3. Sphenopalatine artery (from maxillary artery).



Arterial supply of nose. (continued).

4. Greater palatine artery (from maxillary artery).

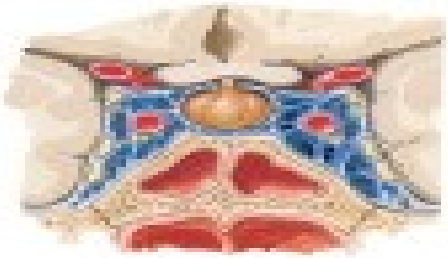
5. superior labial artery (from facial artery).



Venous supply of nose.

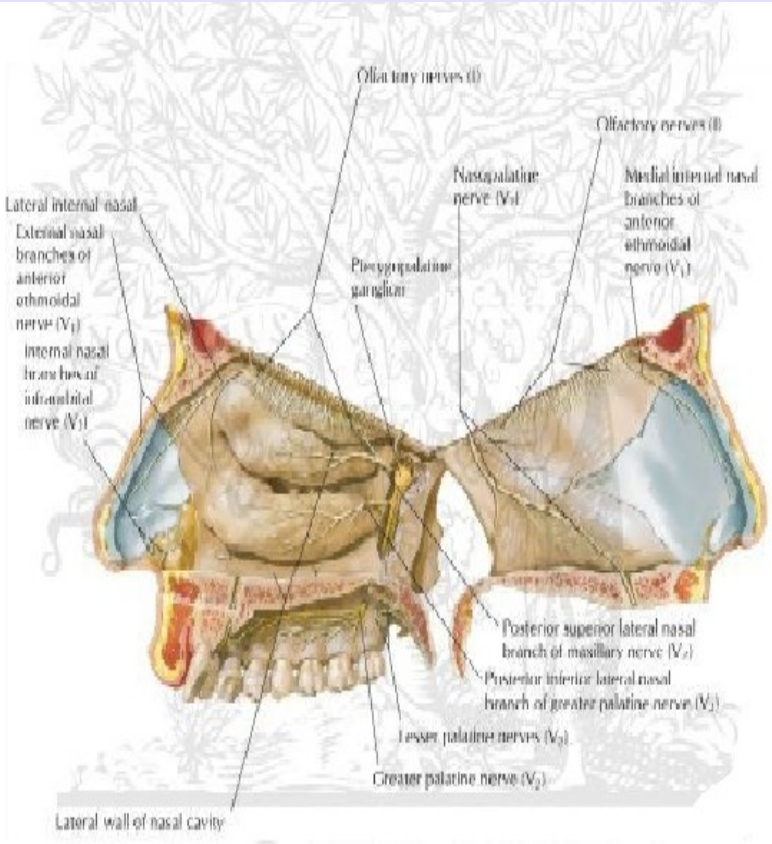
- Submucosal venous plexus.





Nerve supply of nose.

- Sensory – (Trigeminal nerve).
- **Anterior superior** – Anterior-ethmoidal nerve (ophthalmic nerve).
- **Anterior inferior** – Anterior superior alveolar nerve (maxillary nerve).
- **Posterior superior** – Posterior superior, lateral nasal (maxillary nerve).


Olfactory nerves (I)
Olfactory nerves (I)
Nasopalatine nerve (V2)
Medial internal nasal branches of anterior ethmoidal nerve (V1)
Lateral internal nasal
External nasal branches of anterior ethmoidal nerve (V1)
Internal nasal branches of infraorbital nerve (V2)
Pterygopalatine ganglion
Posterior superior lateral nasal branch of maxillary nerve (V2)
Posterior inferior lateral nasal branch of greater palatine nerve (V2)
Lesser palatine nerves (V2)
Greater palatine nerve (V2)
Lateral wall of nasal cavity

Self Assessment.

- Cartilaginous framework of nose is formed by?
- Name the Boundaries of Medial wall (nasal septum)?
- Name the arteries supplying the nose?

THANK YOU ALL.